ADVANCED PROGRAMING

Class Test 1
Advanced Programming
Course Code:CSE 341
Time: 30 minutes Marks:7.5

1. A university wants to develop a Java program to manage information about students who receive scholarships. Create a class named `Person` with an attribute `name`, a constructor for initialization, and a `displayInfo()` method to show basic information. Create a class named `Student` that extends `Person` and adds `studentId` and `marks`. Override `displayInfo()` to display the student's information, ID, and marks. Then create a class named `ScholarshipStudent` that extends `Student` and adds `tuitionFee`. Override `displayInfo()` again to display the complete student information. Implement a `calculateScholarship()` method using these rules: marks $\geq$ 80: 50% of tuition fee; marks 70–79: 30%; marks 60–69: 20%; below 60: no scholarship. In `main()`, create a `ScholarshipStudent` object and assign it to a `Person` reference using upcasting. Invoke `displayInfo()` through the parent reference to demonstrate dynamic method dispatch and late binding. Finally, display the calculated scholarship amount.

Bangladesh University of Business and Technology (BUBT)
Department of Computer Science and Engineering

CT-1 Examination: Fall 2026
Course Code: CSE 341 | Course Title: Advanced Programming
Intake: 55 | Program: B.Sc. Engg. in CSE

Time: 30 Minutes **Marks: 10**

Answer all the questions

CO1: Apply the fundamentals of OOP features alongside advanced programming concepts in java.

[CO1, PO1] **1. (a)** Complete the following Java program by filling in the blanks based on the given hints. Ensure that the completed program correctly displays the online customer's name, ID, and item count. **[5]**

```
class Customer {
  ____ String customerName;
  // Accessible from any class
  ____ int customerId;
  // Accessible only within Customer

  Customer(String name, int id) {
    this.customerName = ____;
    this.customerId = id;
  }

  class ____ {
    private int itemCount;

    Order(int count) {
      this.itemCount = ____;
    }

    void display() {
      System.out.println(customerName +
          " " + customerId + " " + ____);
    }
  }
```

```
void showCustomer() {
  ____ item = new ____(4);
  item.display();
}
}

public class Main {
  public static void main(String[] args) {
    Customer obj = new Customer("Arif", 701);
    obj.____();
  }
}
```

(b) A online store organization wants to evaluate the performance of its customers. Create a Java program using two different packages, store and management. In the store package, create a Customer class with three private data members: customerName, customerId, and loyaltyScore. Use appropriate getter and setter methods to access and modify these data members. The loyaltyScore must be validated to accept only values between 0 and 100. Implement a method calculateLoyaltyTier() to classify the score as Platinum (80-100), Gold (60-79), Silver (40-59), or Bronze (below 40). In the management package, create a CustomerManagement class containing the main() method, create objects for three customers, assign their information using setter methods, and display each customer's name, ID, score, and category. **[5]**

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Time: 30 Minutes **Marks: 10**

Answer all the questions

CO1: Apply the fundamentals of OOP features alongside advanced programming concepts in java.

1, PO1] **1. (a)** Complete the following Java program by filling in the blanks based on the given hints. Ensure that the completed program correctly displays the sports academy's name, ID, and training hours. **[5]**

```
class Athlete {
  ____ String athleteName;
  // Accessible from any class
  ____ int athleteId;
  // Accessible only within Athlete

  Athlete(String name, int id) {
    this.athleteName = ____;
    this.athleteId = id;
  }

  class ____ {
    private int trainingHours;

    Training(int hours) {
      this.trainingHours = ____;
    }

    void display() {
      System.out.println(athleteName +
          " " + athleteId + " " + ____);
    }
  }
```

```
void showAthlete() {
  ____ item = new ____(20);
  item.display();
}
}

public class Main {
  public static void main(String[] args) {
    Athlete obj = new Athlete("Sakib", 501);
    obj.____();
  }
}
```

(b) A sports academy wants to evaluate the training performance of its athletes. Create a Java program using two different packages, academy and management. In the academy package, create a Athlete class with three private data members: athleteName, athleteId, and performanceScore. Use appropriate getter and setter methods to access and modify these data members. The performanceScore must be validated to accept only values between 0 and 100. Implement a method calculatePerformanceLevel() to classify the result as Elite (80-100), Advanced (60-79), Intermediate (40-59), or Beginner (below 40). In the management package, create a AthleteManagement class containing the main() method, create objects for three athletes, assign their information using setter methods, and display each athlete's name, ID, score, and performance level. **[5]**

Bangladesh University of Business and Technology (BUBT)

Department of Computer Science and Engineering

Course Code: CSE 341 | Course Title: Advanced Programming Language

Intake: 55 | Program: B.Sc. Engg. in CSE

Build a Java program named MainOverloadDemo that contains the standard public static void main(String[] args) along with two overloaded main methods having different parameter lists (e.g., main(int x) and main(String a, String b)), where the standard main explicitly invokes the overloaded versions; then determine which main the JVM calls at startup, explain why the overloaded ones are not invoked automatically, analyze what happens if you change the return type of an overloaded main without changing its parameter list, and demonstrate with working code how a command-line utility could use overloaded main methods to separate JVM-triggered execution from internally-triggered batch mode. [6] CO1

Build a payment system where Payment class is the base class with a method processPayment(double amount). Create subclasses CreditCard, PayPal, and Crypto. Demonstrate runtime polymorphism by storing all payment types in a single array and processing a payment of $500 through each. Show how the correct method is called without using if-else or switch. remove you are [4] CO1

Create a base class Appliance with a method turnOn() and encapsulated fields brand and powerRating, then create subclasses WashingMachine, Refrigerator, and Microwave that override turnOn(), and demonstrate runtime polymorphism. Then create a class SmartWasher that extends both WashingMachine and another class SmartDevice (with method connectToWiFi()) to prove that Java does not allow multiple inheritance through classes and explain why the compiler rejects it. [5] CO1

Bangladesh University of Business and Technology (BUBT)
Department of Computer Science and Engineering
Class Test-1: Fall 2026
Course Code: CSE 341 | Course Title: Advanced Programming
Intake: 55, Sec-7, Program: B.Sc. Engg. in CSE

Time– 25 minutes | Marks – 15

1 Develop a banking application containing a class **Account** with **accountNumber**, **accountHolder**, and **balance**. Use a constructor to initialize the account. Overload **deposit()** and **withdraw()** methods to allow depositing or withdrawing either an integer amount or an amount with a transaction description. Demonstrate the system with multiple accounts. 7.5

2 Create a superclass **LibraryItem** containing **itemId**, **title**, and **author**. Create subclasses **Book** and **Magazine**. Use constructors and super to initialize common attributes. Override **calculateFine(int lateDays)** so that books and magazines have different fine rates. Implement a **display()** method to show the information. Create objects of both subclasses and demonstrate the system. 7.5

CT-1

Q1. A university wants to develop a small application that can run on different operating systems without changing the source code. The application should also provide security by preventing direct access to sensitive data and should support code reuse through classes and objects. Based on these requirements, identify three features of Java that make it suitable for developing this application and briefly explain how each feature helps in this scenario. [5 Marks]

Q2. A university email system has different types of users, such as professors and students. Each type of user should send emails differently. Design a Java program using method overriding and demonstrate dynamic method dispatch when sending an email from a professor account. [5 Marks]

Q3. A food delivery system needs to calculate delivery charges in different ways depending on the information provided. Using method overloading, design methods to calculate the charge based on distance only, distance and order weight, and distance, order weight, and premium membership status. Demonstrate the overloaded methods from the main() method. Also state whether the main() method can be overloaded in Java. [5 Marks]

DATABASE SYSTEMS

Bangladesh University of Business and Technology
Department of Computer Science and Engineering

Course Title: Database Systems
Time: 30 Minutes **Set: A** Course Code: CSE 207
Marks: 15

[CLO1] **1.** A university is developing an online course registration system that will be accessed simultaneously by thousands of students through web browsers. **Apply the concepts of two-tier and three-tier database application architectures** to design the system. Identify the components involved in each architecture and show how the client, application server, and database server would communicate. Based on the given scenario, select the more appropriate architecture and justify your choice. [5]

[CLO1] **2.** **Consider the following relational schema:** **[5]**

Course (course_id, course_name, category, start_date, instructor_id)
Instructor (i_id, i_name, total_courses)
Student (s_id, s_name, department)
Course_Student (course_id, s_id)

a) Create the Course and Course_Student tables with appropriate Primary Key and Foreign Key constraints.
b) Find the names of instructors who have taught courses belonging to the "Programming" category.
c) Write an SQL query to find the number of students enrolled in each course. Display the course_id, course_name, and the total number of enrolled students.
d) Write an SQL query **to** find the categories that contain more than 2 courses. Display the category name and the number of courses in each category.
e) Write an SQL query to find the instructor(s) who have taught the highest number of courses. Display the instructor's name and the number of courses taught.

[CLO1] **2.** Using the relational schema given in Question 2, construct appropriate **relational algebra expressions** for the following: [5]

a) Relational Algebra Write a relational algebra expression to display the course name and instructor name for every course.
b) Write a relational algebra expression to find the names of instructors who teach courses in the "Database" category.
c) Write a relational algebra expression to retrieve the names of students, course names, and instructor names for all course enrollments.
d) Using extended relational algebra, write an expression to find the number of students enrolled in each course. Display the course_id, course_name, and number of students enrolled.
(e) Find the course categories that have more than 3 courses. Display each category and its number of courses.

Bangladesh University of Business and Technology
Department of Computer Science and Engineering

Course Title: Database Systems | Course Code: CSE 207

Time: 30 Minutes | **Set: B** | **Marks: 15**

[CLO1] **1.** Apply DDL and DML commands to perform database operations, and demonstrate the differences between DDL and DML with appropriate SQL examples.

[CLO1] **2.** **Consider the following relational schema:**

Doctor (doc_id, doc_name, specialization, joining_date, hospital_id)
Hospital (h_id, h_name, total_doctors)
Patient (p_id, p_name, city)
Doctor_Patient (doc_id, p_id)

a) Create the Doctor and Doctor_Patient tables with appropriate Primary Key and Foreign Key constraints.
b) Find the hospitals that have more than 5 doctors. Display the hospital name and the total number of doctors.
c) Find the number of patients treated by each doctor. Display the doctor's name and the number of patients. Include doctors who have not treated any patients.
d) Find the names of doctors who work in the hospital with the highest number of doctors. Display the doctor's name and specialization.
e) Find the total number of doctors working in each hospital. Display the hospital name and the number of doctors.

[CLO1] **2.** Using the relational schema given in Question 2, construct appropriate **relational algebra expressions** for the following: [5]

a) Find the names and specializations of all doctors whose specialization is 'Cardiology'.
b) Find the names of doctors and their specializations who have treated patients from 'Chittagong'.
c) Find the names of doctors who have treated patients from 'Dhaka'.
d) Find the hospital name and the number of doctors working in each hospital, and display only those hospitals where the number of doctors is greater than 5.
e) Find the names of doctors who work at the hospital having the maximum number of doctors.

ıgladesh University of Business and Technology (BUBT)

Department of Computer Science and Engineering

ırse code: CSE 207 **Course title: Database Systems**

ne: 45 min Marks: 15

1. Suppose a university stores student records, course registrations and fee payments in separate files. The staff report these problems:

- **Problem-1:** A student's phone number is different in the registration file and the fee file.
- **Problem-2:** The system crashes after the fee is deducted but before the registration is saved.
- **Problem-3:** Every employee can open every file, including grades.
- **Problem-4:** Two clerks update the same student's balance at the same time and the final value is wrong.

Apply the concept of the purpose of database systems and name the file-system drawback shown by each problem, and briefly explain how a database system solves it. [4]

2. Suppose a university employee submits the query: **select name from instructor where dept_name = 'Physics'**. Apply the concept of query processing and briefly explain the role of each step in this scenario. [5]

3. Suppose a university wants to store **student(ID, name, dept_name, tot_cred).**

(a) Construct a create table statement for student. ID must be the primary key, name must never be empty, and tot_cred must not be negative.

(b) Three users need these results from the table:

- **User-1:** All students whose name starts with "A", in alphabetical order of name.
- **User-2:** All students whose name contains "an" anywhere.
- **User-3:** All students ID, name and dept_name, sorted by dept_name and then by name.

Develop an SQL query for each user. [6]

Bangladesh University of Business & Technology (BU

Department of Computer Science and Engineering

Semester: Fall 2026 **Sub: CSE 207(Database System)**

Class test 01 Intake: 55/6 Mark: 15

Q1. **Explain** the various application of database management system.	**[4]**
Q2. **Explain** the integrity constraints are initialized through DDL.	**[4]**
Q3. **Explain** the keys used in database and their purpose.	**[4]**
Q4. Draw a schema diagram for the following database: R1(a1, a2, a3, c1) R3(b1, b2, a1) R3(c1, c2, b1) R4(b1, c1, d1)	**[3]**

Bangladesh University of Business and Technology (BUBT)

Course Title: Database Systems Course Code: CSE 207 Class Test-1

Time: 40 Minutes (with submission)

1.	Consider the following relational schema: Employee(emp_no, name, address, dept, salary) Books(isbn, title, authors, publisher) Loan(emp_no, isbn, date) Write SQL queries to accomplish these tasks: i. Calculate the total number of employees and then rename the column name with "Total no. of employees". ii. Find those books which has a "s" anywhere in it's title. iii. Display the employee names and the ISBNs of the books they borrowed. iv. Suppose an employee is currently living in **Mirpur 2**, and you need to update their address to **Mirpur 10**. Write the SQL query to perform this update in the Employee table.	[4]
2.	Express Q1 (i,iii) using appropriate relational algebra operations.	[2]
3.	Given the relations above, identify a suitable primary key and foreign key.	[1.5]

OPERATING SYSTEMS

Class Test 1: Fall 2026
Course Code: CSE 209 | Course Title: Operating Systems
Intake: 55, Program: B.Sc. in CSE

Time– 35 Minutes **Marks – 15**

1. **(a)** A computer system uses dual-mode operation to protect system resources and ensure secure execution of programs. The operating system distinguishes between user mode and kernel mode to control access to privileged instructions and hardware resources. **Illustrate** the working mechanism of dual-mode operation, highlighting how the system switches between user mode and kernel mode during program execution. **[5]**

2. Imagine a series of processes, labeled P1 through P5, that are scheduled for execution. These processes arrive for execution at various times, as indicated in the table below. **[10]**

Process	Arrival Time	Burst Time	Priority
P1	0	9	2
P2	2	4	12
P3	5	10	10
P4	6	6	6
P5	8	2	4

Apply the FCFS and Preemptive SJF scheduling to find the average turnaround time, average waiting time and average response time.

Bangladesh University of Business and Technology (BUBT)

Department of Computer Science and Engineering

Quiz Examination: Fall 2026 | Course Code: CSE 209 | Course Title: Operating Systems

Intake: 55 | Program: B.Sc. Engg. in CSE

Time: 30 Minutes **Marks: 15**

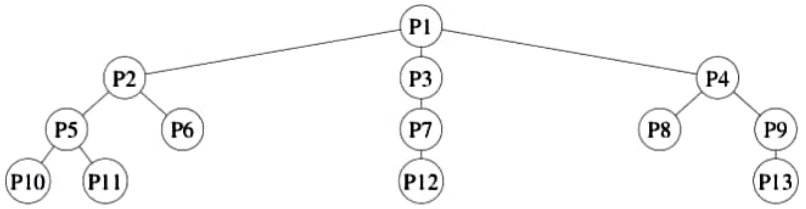
(Answer all the questions)

CO1: Apply fundamental operating system concepts and structures to interpret the behavior of processes, and resources in a computing system.

[CO1, PO1] **1.** Consider the process tree shown below. Suppose processes **P2** and **P4** get terminated. List the PIDs of the *orphan* processes in each of the following cases (write "None" if there are no orphans): **[4]**

(a) A system that supports **cascading delete**.

(b) A system that does **not** support cascading delete.



(a) ..... (b) .....

[CO1, PO1] **2.** Briefly explain the **shared memory communication** of a process with a suitable diagram. **[3]**

[CO1, PO1] **3.** Define the following terms: (i) Timer, (ii) Multiprogramming, (iii) Unbounded buffer, (iv) Marshalling. **[8]**

CT - 1

Operating System

CSE - 209

Time: 50mins, Marks: 15

1. A computer system is running three processes, P1, P2, and P3, in the ready queue. The CPU scheduler selects P1 for execution, but P1 requests a print operation and must wait for it to complete. Thus, the OS moves P1 to the other queue and allocates the CPU to P2. After the print operation finishes, P1 returns to the ready queue. **Draw** a diagram showing how P1 moves between these queues and **explain** the role of each queue. **[5]**

2. What's RPC? Explain the procedure of RPC? **[5]**

3. How do cooperating processes exchange information to complete tasks using different interprocess communication (IPC) frameworks? **[5]**

CSE 209 Operating System - Ct1 [CO1] – 3*5=15

1. Compare STS, LTS and MTS in terms of where and how they are used. Which of these controls the degree of multiprogramming? How?

2. In a shared computing system, multiple users can execute their tasks. Between the multiprogramming and multitasking approaches, which one is more suitable? Justify your answer.

3. Make a comparison between RPC and interprocess communication. Highlight their differences in project work, how evaluation matters, and present with a diagram.

Remote Procedural Call

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CSE 209 Operating System - Ct1 [CO1] – 3*5=15

1. Describe the activities performed by the OS to aid in process as well as memory management

2. Suppose the CPU needs to switch between processes A and B to ensure a multitasking environment. Explain how the switching is done, with a diagram. What is the role of the PCB here?

3. In RPC, the sender system does not know the port number of the process it wants to communicate with. Explain with a diagram how this issue is resolved.

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